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Section 1

Game Overview

Concept & Elevator Pitch

PawSanctuary is a merge-puzzle idle casual game for iOS where players build and manage an animal rescue sanctuary. Players spend Kibble (energy) to spawn animals, merge identical animals to advance them through a 15-stage chain, and unlock new animal families as they progress. Each family also spawns themed sub-objects that merge into powerful consumables.

Elevator pitch: Run an animal sanctuary, rescuing creatures across 15 families and merging them from Infant through 15 evolutionary stages to Legendary. Every family produces themed sub-objects that merge into board-altering power-ups, while a rich system of daily challenges, weekly spotlights, and milestone titles keeps players coming back for months.

Genre	Merge puzzle / idle casual
Platform	iOS (iPhone primary, iPad secondary)
Target Audience	Casual mobile gamers, animal lovers, ages 18–45
Monetization	Free-to-play with IAP and optional subscription
Comparable Titles	Travel Town, Tasty Travels, Merge Mansion
Differentiator	15 named animal families with emotional attachment; sub-object buff system; Ambassador milestone

Design Pillars

- **Warmth** — every interaction feels like caring for animals, not grinding.
- **Value** — players always feel the game is generous compared to competitors (30–40% cheaper per Kibble than Travel Town).
- **Momentum** — there is always a next thing to do; dead ends are eliminated by design.
- **Clarity** — new players understand the core loop within 60 seconds.

Core Loop Summary

Phase	Action	Resource
Open app	Claim daily login reward	Kibble / Dog Tags
Session start	Check timed rescue requests	Urgency timer
Active play	Spend Kibble to rescue animals	Kibble
Active play	Merge matching animals to advance through 15 stages	Board space
Active play	Merge sub-objects to earn consumables	Sub-object drops
Active play	Complete daily challenges and quests	XP / rewards

Session end	Store excess animals in inventory	Inventory slots
Exit	Return when Kibble regenerates or notification fires	Regen / push

Section 2

Core Mechanics

Board Configuration

The game board is a **9×7 grid = 63 cells**. Two rows (rows 7 and 8) begin locked and unlock at player levels 3 and 8 respectively. This supersedes any earlier reference to a 6×5 board — the 9×7 implementation is authoritative.

Merge Mechanic

Scenario	Result
Same family + same stage	Merge → next stage (tap-to-merge / drag)
Different family OR different stage	Swap positions
Source → empty cell	Move to empty cell
Drag off bottom of board	Send to inventory (Merge tab)
Two items at max tier (stage 15)	Ambassador celebration; item enters Sanctuary Collection

Energy (Kibble) System

Property	Value
Regen rate	1 Kibble per minute
Regen cap (base)	100 Kibble
Regen cap (level 10+)	150 Kibble (expanded per F.Q2 decision)
Cost per rescue (spawn)	1 Kibble
Auto-spawn after merge	Free
Cap removal	Unlimited via IAP purchases

Spawner Mechanics

Each animal family has a **family spawner** — a persistent tile on the board that costs Kibble to activate. The family spawner has no charge limit and produces animals at a tier determined by its spawn multiplier. Legacy producer tiles exist for earlier content but the family spawner is the primary mechanic.

Tap-select UX: Tapping a spawner selects it (highlighted border). A second tap on the same spawner activates it (costs Kibble, produces output). Tapping elsewhere deselects. Power-up consumables are applied by selecting a power-up from inventory, then tapping a spawner.

Sub-Object Drop System

Every family spawner activation has a **20% base chance** to produce a sub-object instead of an animal. Within that 20%, the sub-object rarity is determined by:

Rarity	Effect	Drop Weight
Common — Speed Burst	2× spawner production speed for 30 seconds	60%
Uncommon — Map Supplies	Rare building resource for area/map expansion	25%
Rare — Spawner Refill	Instantly restores all production charges	10%
Epic — High-Tier Drop	Forces next spawner output to stage 2 or 3	5%

The 80% of activations not generating a sub-object always produce an animal. Pity timers (30 spawns → Rare guarantee; 60 spawns → Epic guarantee) apply per family.

Power-Up Application

Power-ups are produced when a sub-object reaches stage 4 (the consumable tier). They route to the power-up inventory. To apply: tap to select a power-up from inventory, then tap a target spawner. Visual feedback (glow / badge) confirms the buff is active on the spawner tile.

Sell Animal Mechanic

Tier (Stage)	Sell Value (Coins)
1	1
2	2
3	5
4	10
5	20
6	50
7	100
8	200
9	500
10	1,000
11	2,000
12	5,000
13	10,000
14	25,000
15 (Ambassador)	100,000

Section 3

Animal System

15 Animal Families

Family	Family Group Name	Starter?
Canines	Dog family	Yes
Felines	Cat family	Yes
Avians	Bird family	Yes
Rodents	Hamster/rodent family	Yes
Aquatics	Fish family	Yes
Ursids	Bear family	No — stage 6 gate
Cervids	Deer family	No — stage 6 gate
Equines	Horse family	No — stage 9 gate
Bovines	Cattle family	No — stage 9 gate
Reptiles	Reptile family	No — stage 9 gate
Amphibians	Frog/amphibian family	No — stage 12 gate
Primates	Primate family	No — stage 12 gate
Lagomorphs	Rabbit family	No — stage 12 gate
Pachyderms	Elephant/pachyderm family	No — stage 14 gate
Marsupials	Kangaroo/marsupial family	No — stage 14 gate

15-Tier Chain Structure

Each family has exactly 15 merge stages arranged into 5 Eras of 3 stages each. Reaching stage 15 (the Legendary tier) triggers the Ambassador celebration and removes the animal to the Sanctuary Collection.

Era	Stages	Theme	Visual Treatment
Era 1 — Infant/Juvenile	1–3	Baby animals, just arrived	Desaturated, small silhouette
Era 2 — Adolescent	4–6	Young but growing	Partial color, medium size
Era 3 — Young Adult	7–9	Full-grown variants	Full color
Era 4 — Adult	10–12	Powerful, distinguished	Golden border accent
Era 5 — Legendary	13–15	Mythic / ultimate forms	Animated glow, star badge

Sample Chain: Canines

Stage	Name	Era
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1	Pup	Infant
2	Kit	Infant
3	Houndling	Infant
4	Terrier	Adolescent
5	Spaniel	Adolescent
6	Scout	Adolescent
7	Retriever	Young Adult
8	Shepherd	Young Adult
9	Husky	Young Adult
10	Alpha	Adult
11	Guardian	Adult
12	Sentinel	Adult
13	Dire Wolf	Legendary
14	Mythic	Legendary
15	Primordial	Legendary

Ambassador System

When any animal reaches stage 15 (Legendary), it is removed from the board and becomes an **Ambassador**. The ambassador counter increments, triggering MilestoneManager at thresholds. The ambassador counter drives a standing repeatable quest (collect 3 Ambassadors → 500 coins).

Sub-Object Chains

Each family also has a **4-stage sub-object chain**. Stage 4 is the consumable power-up. Sub-objects drop from family spawners (20% base rate) and merge exactly like animals but cannot merge with animals. There are 60 unique sub-object items total (15 families × 4 stages).

Family	Stage 1	Stage 2	Stage 3	Stage 4 (Power-Up)
Canines	Biscuit	Bone	Chew Toy	Golden Ball
Felines	Bell	Feather Wand	Yarn Ball	Laser Pointer
Rodents	Seed	Nut	Berry	Corn Cob
Avians	Down	Plume	Quill	Iridescent Tail
Bovines	Clover	Hay Bale	Salt Lick	Water Trough
Equines	Curry Comb	Brush	Sponge	Trophy
Ursids	Honey Comb	Salmon	Wild Hive	Berry Bush
Cervids	Leaf	Sprout	Twig	Antler
Aquatics	Shell	Pearl	Starfish	Treasure Chest

Reptiles	Pebble	Sand	Warm Moss	Heat Lamp
Amphibians	Duckweed	Reeds	Lotus Flower	Algae Bloom
Primates	Banana	Mango	Papaya	Jungle Vine
Pachyderms	Puddle	Mud Clump	Clay Mound	Rainfall
Lagomorphs	Lettuce	Carrot	Cabbage	Turnip
Marsupials	Bud	Flower	Leaf Bundle	Bark

Section 4

Progression Systems

Player Levels & XP

Players earn XP from merges, quest completion, and login streaks. Level-ups unlock cosmetic rewards (board themes, animal name tags) and expand gameplay (level 3 and level 8 unlock additional board rows). Player level also gates the maximum tier that adoption orders can request, with tiers 0–14 accessible progressively as level increases.

Milestone Titles

Ambassador Count	Title	Reward
5 Ambassadors	Junior Rescuer	10 Kibble bonus
15 Ambassadors	Animal Friend	5 Dog Tags
30 Ambassadors	Sanctuary Guardian	Free inventory tab unlock
50 Ambassadors	Legendary Rescuer	Exclusive rare family unlock

Daily Login Rewards

Day	Kibble	Dog Tags
Day 1	+5	—
Day 2	+10	—
Day 3	+5	+2
Day 4	+15	—
Day 5	+10	+3
Day 6	+20	—
Day 7 (streak bonus)	+15	+5

Cycle repeats after Day 7. Missing a day resets streak to Day 1.

Quest / Order System

The Adoption Board maintains 2 active time-limited rescue orders at all times, each with a 10-minute countdown. Orders specify a target animal family and tier; completing an order awards 5–10 Kibble and 1–3 Dog Tags. Orders auto-replace on expiry. A skip mechanic (costs Dog Tags) lets players replace an undesirable order. The order difficulty is weighted by player level, mapping to achievable tiers.

Pity Timer System

Per-family spawn counters track how many activations have occurred without a Rare or Epic sub-object drop. At **30 spawns** without a Rare, the next sub-object drop is guaranteed to be Rare or better. At **60 spawns** without an Epic, the next sub-object drop is guaranteed Epic. These counters are stored per-family in GameState and persist across sessions.

Weekly Family Spotlight

One family is featured each calendar week (rotating Monday midnight). All merges of the spotlight family grant 2× score. A weekly milestone (merge 10 spotlight animals) awards +20 Kibble and +5 Dog Tags. A golden border is shown on board cells containing the spotlight family.

Daily Challenges

Difficulty	Example Goals	Completion Reward
Easy	Merge 3 animals; Rescue 4 strays; Merge 2 of family X	+10 Kibble, +2 Dog Tags (all 3)
Medium	Merge 6 animals; Get 1 animal to Era 3; Rescue 5 strays	+25 Kibble, +5 Dog Tags; 7-day streak: +5 Kibble
Hard	Get 1 animal to Era 4; Merge 10 animals; Reach Stage 10	+35 Kibble, +8 Dog Tags

Section 5

Map & Areas

SanctuaryMap — Current Areas (6 Live)

Area Name	Family Unlocked	Status
Antique Dog House	Canines	Tutorial — always unlocked
Scratching Post	Felines	Live
Garden Hutch	Lagomorphs	Live
Decorative Birdhouse	Avians	Live
Wooden Burrow	Rodents	Live
Heated Rock	Reptiles	Live

Each area costs tier-5 material combinations and has 2 upgrade tiers granting UpgradeBonus passives.

Phase 5 — 9 Remaining Families

Nine new SanctuaryArea entries will unlock the remaining families. Build costs scale logarithmically to reflect late-game investment. The Aquatics milestone unlock path (previously tied to 50 ambassador completions) is retired; Aquatics is unlocked via its map area purchase.

Family	Notes
Ursids (Bears)	New area unlock
Aquatics (Fish)	Map area replaces milestone path (F.Q4 decision)
Amphibians	New area unlock
Marsupials	New area unlock
Primates	New area unlock
Equines (Horses)	New area unlock
Pachyderms	New area unlock
Bovines (Cattle)	New area unlock
Cervids (Deer)	New area unlock

Area Upgrade Bonuses

Bonus Type	Effect
Regen Rate	Increases Kibble regen rate for this area's family
Spawn Rate	Reduces spawner cooldown
Order Frequency	Increases rate of adoption orders for this family

Sub-Object Drop Rate	Increases the 20% base sub-object drop rate (new — Phase 5)
Pity Timer Reduction	Reduces spawn count required for Rare/Epic guarantee (new — Phase 5)
Power-Up Duration	Extends active power-up duration (new — Phase 5)

Build Cost Materials

Area construction requires wood, stone, and crystal materials in tiered quantities. Materials exist in 6 tiers (1–6). Current areas cost tier-5 combinations. Phase 5 areas will cost tier-5 and tier-6 materials in larger quantities, with some areas requiring cross-type mixes (e.g., wood tier 6 + crystal tier 4). The Building Supplies inventory tab (limitless, auto-merge) holds all build materials without slot pressure.

Section 6

Economy & Currencies

Currency	Type	Primary Source	Primary Sink
Kibble	Energy (soft)	Timer regen (1/min), quests, login	Rescue spawns (1 each), IAP
Coins	Soft	Sell animals, orders, quests	Future: board upgrades
Dog Tags	Premium soft	Quests, challenges, milestones	Inventory unlocks, skips
Gems	Hard	IAP	Kibble top-up, premium features
Sanctuary Pass	Subscription	\$4.99/month	+30 Kibble/day + perks

IAP Products

Product ID	Name	Contents	Price
kibble.small	Small Kibble Bag	60 Kibble	\$0.99
kibble.medium	Medium Kibble Bag	180 Kibble	\$1.99
kibble.large	Large Kibble Bag	600 Kibble	\$4.99
dogtags.small	Dog Tag Pack	15 Tags	\$0.99
dogtags.medium	Dog Tag Bundle	60 Tags	\$2.99
dogtags.large	Dog Tag Jackpot	175 Tags	\$6.99
bundle.starter	Sanctuary Starter Pack	100 Kibble + 20 Tags	\$1.99
pass.monthly	Sanctuary Pass	+30 Kibble/day + perks	\$4.99/mo

Sanctuary Pass Perks

- +30 Kibble bonus on first daily open
- 1 free timed rescue request skip per day
- 1 free Spawner Refill per day
- Exclusive 'Pass' badge on profile (future)
- Early access to seasonal events (future)

Value Positioning vs. Competitors

Travel Town equivalent: ~\$1.99 for ~20 minutes of energy. PawSanctuary target: \$1.99 for ~60 minutes of energy (3× better value). Large bundle per-unit cost is always 40%+ cheaper than the equivalent small bundle. The Sanctuary Pass daily bonus is recoverable in ~3.3 days at full play.

Section 7

Card System

The card collection mechanic (implemented in **CardSystem.swift**) enables players to collect, trade, and display cards representing their sanctuary animals. Cards are awarded through gameplay milestones and can be acquired via Card Packs.

Card Packs

Card packs contain a randomized set of animal cards with rarity tiers matching the animal era system (Infant through Legendary). Packs are purchasable with Dog Tags or earned via weekly milestone completion.

Album View

An album view displays all collected cards organized by family. Completing a family's card set awards a permanent bonus for that family. The star count (total cards collected / total possible) integrates with the progression display and feeds into the leaderboard system.

Trading

Players can offer duplicate cards for trade via the Game Center social layer. The trading mechanic encourages friend connections and repeat engagement with the card system. Trading is planned for Phase 4.

Section 8

Social & Competitive

Game Center Integration

PawSanctuary integrates Game Center for leaderboards and achievements via the existing authenticateGameCenter flow in MergeBoardViewModel. The InviteSystem provides friend discovery infrastructure.

Planned Social Features

Feature	Description	Status
Weekly Leaderboard	Ranked by ambassador count; resets Sunday midnight	Planned — Q4 2026
Community Rescue Events	Cooperative goal (e.g., 10,000 rescues); all players participate	Planned — Q1 2027
Friend Gifting	1 free kibble gift/day per friend (adds 5 kibble to recipient)	Planned — Q4 2026
InviteSystem	Game Center friend discovery for gifting and leaderboards	Infrastructure live

Section 9

Notifications & Retention

Trigger	Message	Timing
Kibble full (cap reached)	"Your Kibble bag is full! Come rescue some animals!"	Immediate
Daily challenge reset	"New daily challenges are ready at the sanctuary!"	Midnight
Rescue order expiring	"Your rescue request expires in 2 minutes!"	T-2 min
Login streak at risk	"Don't break your streak! Log in before midnight."	8 PM local
Weekly spotlight change	"New spotlight family this week: [Name]!"	Monday midnight
Spawner idle	"Your [Family] Spawner has run out of charges!"	On idle

Push Notifications capability must be enabled in Xcode (target → Signing & Capabilities). SoundManager.swift and HapticManager.swift must be added to the Xcode project target manually.

Seasonal Events (Planned)

Four seasonal events per year: Spring Rescue, Summer Safari, Autumn Harvest, Winter Warmth. Each runs 7 days and features exclusive rare families/stages, event currency earned through special merges, and a leaderboard showing top rescuers during the event window.

Section 10

Audio & Haptics

SoundManager (SoundManager.swift)

Sound Event	Trigger
merge	Two matching animals merge successfully
cell_unlock	A new board cell is unlocked
quest_claim	Player claims a quest reward
daily_challenge	Player completes a daily challenge
rescue_claim	Player claims a rescue request bonus
button_tap	General UI button interaction

Current implementation uses `AudioServicesPlaySystemSound`. Replace with `AVAudioPlayer` instances pointing at final audio asset files when assets are ready.

HapticManager (HapticManager.swift)

Haptic Intensity	Trigger
Light	Board tap (tile selection)
Medium	Successful merge
Success (notification)	Board cell unlock, quest claim, daily challenge completion

Mute Toggle

A mute toggle in the game settings persists the player's audio preference via `UserDefaults`. The toggle controls both sound effects and background music (when music is implemented). Haptics respect the iOS system-level haptics toggle.

Section 11

Technical Architecture

State Management

PawSanctuary uses **SwiftUI with @Observable** for all view models. The app moved from ObservableObject to the modern @Observable macro pattern; SoundManager.swift is the only remaining class using the old pattern (flagged for migration in QA-02). All state mutations happen on @MainActor.

Persistence — Codable Save System

Game state is persisted as JSON via a custom Codable save system. The current save version is **v23**. Versioned migrations (v1 through v23) handle schema evolution. Each migration is additive where possible — new fields are injected with defaults using decodeIfPresent. Save/load occurs on a 5-second auto-save tick and on every user action.

ContentRegistry

ContentRegistry is the single source of truth for all chain data. It holds all 15 animal chains (15 tiers each), 15 sub-object chains (4 tiers each), supply chains, material chains, and toolbox chains. The save never stores display data — only chainID (String) and tier (Int). All names, icons, and colors are looked up from the registry at runtime.

iCloud KV Sync

iCloud Key-Value sync entitlement is provisioned. Activation is pending in Xcode capabilities. KV sync stores the most recent save snapshot for cross-device continuity.

Sub-Coordinators

Coordinator	Responsibility
KibbleEngine	Kibble regen, cap enforcement, cost deduction
QuestCoordinator	Quest generation, progress tracking, completion detection
InventoryStore	Animal / spawner / building-supply slot management
AdoptionBoard	Time-limited rescue order generation and expiry
PlayerProgression	XP, level-up, milestone tracking
SanctuaryMap	Area unlock, build cost validation, upgrade bonus accumulation
MilestoneManager	Ambassador count thresholds; title awards and milestone rewards

Section 12

Known Limitations & Code Health

Source: Code Health Report — June 2026. Fix critical bugs before next TestFlight build.

Critical Bugs (Fix Immediately)

BUG-01 — Duplicate Notification Delegate

Two `NotificationManager` instances are created (the singleton and a second local instance in `MergeBoardView`). Whichever instance initializes last takes the system delegate slot; the other's foreground notification handling is silently non-functional. The result is that some in-app notification presentations may never appear. Fix: remove the local instance in `MergeBoardView` and use the shared singleton throughout.

BUG-02 — Daily Reward Claims Not Persisted on Crash

When a player claims their daily login reward or Sanctuary Pass daily bonus, the granted Kibble and Dog Tags are added to the in-memory game state but the save is not written to disk before the next auto-save tick (5 seconds). A crash within that window causes the player to lose the claimed reward and see it available again on next launch. Fix: call `persist()` immediately at the end of both claim methods.

BUG-03 — Timer Pauses During Scroll Gestures

The main game timer (used for Kibble regen, producer cooldowns, adoption order countdown, and the 5-second auto-save) uses `Timer.scheduledTimer` in the default `RunLoop` mode. When the player scrolls the board or inventory, `UIKit` switches the `RunLoop` to tracking mode, silently pausing all timer ticks. Regen stops, cooldowns freeze, and the auto-save does not fire while scrolling. Fix: switch to `RunLoop.common` mode or use a `DispatchSourceTimer`.

Planned Architectural Improvement: BoardStateManager

The most impactful long-term refactor is extracting board manipulation logic from `MergeBoardViewModel` into a dedicated **BoardStateManager** class. This class would own the board array, cell locking, empty cell lookup, tile placement, merge resolution, and producer tick logic. `MergeBoardViewModel` would become a pure coordinator: it would receive `MergeResult` events and dispatch them to `KibbleEngine`, `QuestCoordinator`, and `PlayerProgression`.

Benefits: eliminates 15+ redundant array traversal calls per interaction, centralizes `recalcBoardIsFull()`, enables isolated unit testing of board logic, and makes performance fixes (PERF-01 through PERF-04) trivial to apply safely. Scheduled as a dedicated sprint after Phase 5 map expansion ships.

End of PawSanctuary Game Design Document v3.0 — June 2026. This document should be updated each time a major feature ships or the design changes.